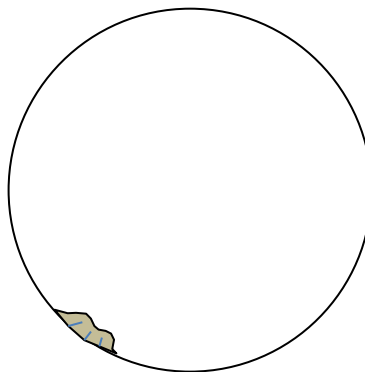


- 9 The diagram shows an object rotating in a vertical circle in the drum of the spin drier which has a constant rate of rotation.



As the drum rotates, the object remains in contact with the surface of the drum.

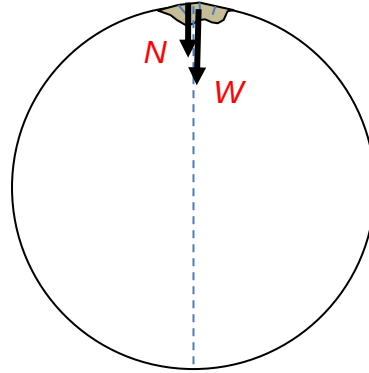
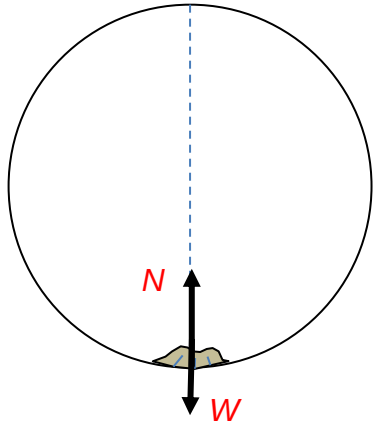
Which statement about the magnitude of contact force on the object is **not** true?

- | | |
|----------|---|
| A | It is always greater than zero. |
| B | It is always equal to the magnitude of the centripetal force. |
| C | It is minimum when the object is the top of the circle. |
| D | It is maximum when the object is the bottom of the circle. |

Answer: B

N - contact force on object, W - Weight of object, F_c - Centripetal force on object

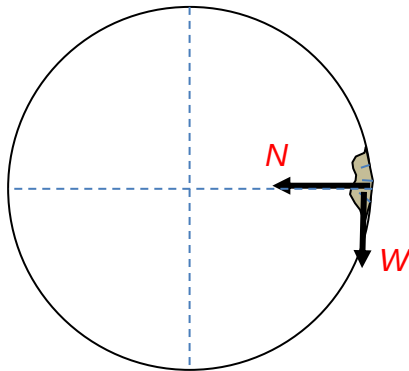
For the object to remain in contact with the surface of the drum, $N > 0$. Option A is correct.



At the bottom and top of circle, the resultant of N and W provides the centripetal force.

At the top of circle, $N + W = F_c$, $N = F_c - W$. This is minimum value of N . Option C is correct.

At the bottom of circle, $N - W = F_c$, $N = F_c + W$. This is maximum value of N . Option D is correct.



At the above position, only N provides the centripetal force, therefore $N = F_c$.

Option B is NOT true.